

Senior Backend-Focused .NET Technical Test

Overview

Build and modernize backend components of an enterprise SaaS platform using C#.NET Core and MSSQL. Focus on clean architecture, scalability, security, GDPR compliance, and testability.

Part A – API Design & Implementation

- Build a REST API for managing Customers and Orders.
- Implement CRUD operations for Customers.
- Implement Order creation linked to a Customer.
- Apply filtering (date range, status) and pagination for Orders.
- Use DTOs (do not expose entities directly).
- Apply clean architecture principles (Controller → Service → Repository).
- Use async/await properly throughout the stack.

Part B – Database & MSSQL Optimization

- Design MSSQL schema for Customers and Orders.
- Add proper indexing strategy for filtering queries.
- Optimize queries to prevent N+1 issues.
- Demonstrate understanding of execution plans.
- Use EF Core properly with performance considerations (AsNoTracking, projection, etc.).

Part C – Security & Compliance

- Implement authentication & authorization (JWT-based).
- Ensure sensitive data handling aligns with GDPR principles.
- Apply input validation and protection against SQL injection, XSS, etc.
- Explain how you would implement audit logging and data retention policies.

Part D – Testing & DevOps Awareness

- Write unit tests for the service layer using a testing framework (e.g., xUnit).
- Write integration tests for API endpoints.
- Demonstrate mocking of dependencies (e.g., Moq).
- Describe how you would integrate tests into CI/CD pipeline.
- Explain how you would containerize the application using Docker.